

Date	03 July 2026
Team ID	SWTID-2026-4593
Team Size	3
Team Leader	Hari Bellamkonda
Team Member 1	Kesha Vardhan Chinni
Team Member 2	Akili Venkata Siva Sai Srikar
Project Name	Machine Learning-Based Credit Card Approval Prediction System
Maximum Marks	4 Marks

Technology Stack

Introduction

The Technology Stack defines the programming languages, frameworks, libraries, databases, development tools, deployment platforms, and supporting technologies used to develop the Machine Learning-Based Credit Card Approval Prediction System. The selected technologies provide high performance, scalability, maintainability, security, and efficient machine learning model deployment.

Layer	Technology	Purpose
Frontend	HTML5, CSS3, Bootstrap 5, JavaScript	Designs responsive user interfaces for applicants and administrators.
Backend	Python, Flask	Implements application logic, API routing, authentication, and prediction services.
Database	MySQL	Stores user accounts, applicant details, prediction history, and system records.
Machine Learning	Scikit-learn, XGBoost, Pandas, NumPy	Supports data preprocessing, feature engineering, model training, evaluation, and prediction.
Data Visualization	Matplotlib, Seaborn	Generates analytical charts and dataset visualizations.
Development Environment	Visual Studio Code, Jupyter Notebook	Used for software development, experimentation, debugging, and model training.

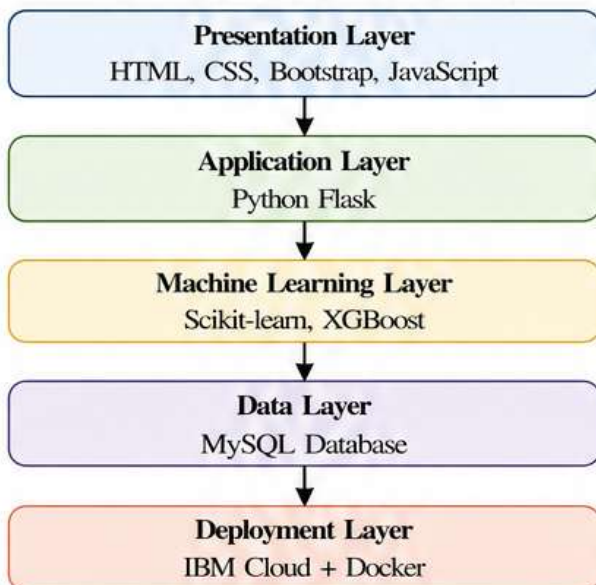
Technology Summary

The selected technologies provide a robust foundation for developing an intelligent, secure, scalable, and maintainable Machine Learning-Based Credit Card Approval Prediction System.

Technology Stack (Continued)

Layer	Technology	Purpose
Cloud Deployment	IBM Cloud, Docker	Deploys the trained Machine Learning model and web application with scalable cloud infrastructure and containerization.
Version Control	Git, GitHub	Tracks source code changes, supports collaboration, manages versions, and maintains project documentation.
Model Storage	Joblib	Stores and loads trained Machine Learning models efficiently for prediction.
API Testing	Postman	Tests REST APIs and validates application endpoints.
Testing	PyTest	Performs functional and integration testing to improve software quality.
Operating System	Windows 11	Provides the primary software development environment.
Virtual Environment	Python venv	Manages isolated project dependencies and package versions.
Documentation	Markdown, Microsoft Word	Used for project documentation, technical reports, and GitHub README preparation.

Software Architecture Summary



Advantages of the Selected Technology Stack

- High prediction accuracy
- Secure application development
- Scalable cloud deployment
- Easy maintenance
- Fast development cycle
- Open-source technologies
- Cross-platform compatibility
- Efficient Machine Learning model integration

Conclusion

The selected technology stack combines modern web technologies, powerful Machine Learning frameworks, reliable database management, cloud deployment services, and collaborative development tools. Together, these technologies enable the Machine Learning-Based Credit Card Approval Prediction System to deliver secure, scalable, maintainable, and accurate prediction services suitable for real-world banking applications.